Bright learn Exercise 03
DATA ANALYTICS SQL Foundations

SQL CASE STATEMENTS

Question1

① SELECT
    Product_name,
    Price
CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN 'Midrange'
    WHEN price < 100 THEN 'Budget'
    END AS price_category ;

TABLE Products

Product_name	Price	Price category
Laptop	1200.00	Expensive
Phone	800.00	Midrange
Keyboard	45.00	Budget
Monitor	200.00	Mid-range
Mouse	25.00	Budget.

Order table Question 2

SELECT
    customer_name,
    amount
CASE WHEN amount >= 1000 THEN 'High value'
    WHEN amount >= 500 THEN 'Medium value'
    WHEN amount < 500 THEN 'Low value'
END AS order_value_category
FROM orders;

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Customer_name	Amount	Order_value_category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value.

The employees table. Question 3

```
SELECT
    emp_name,
    department,
    salary
CASE
    WHEN department = 'IT' AND salary > 80000
        THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55000
    THEN 'Experienced HR'
    ELSE 'staff'
END AS "position_level"
FROM employees;
```

emp_name	department	salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	finance	95000	Staff
Tom	HR	55000	Staff

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The Students Table Question 4

```
SELECT
    student_name,
    score
CASE
    WHEN score >= 90 THEN 'A'
    WHEN score >= 80 THEN 'B'
    WHEN score >= 70 THEN 'C'
    WHEN score >= 60 THEN 'D'
    ELSE 'F'
END AS grade
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

The deliveries table Question 5

```
SELECT
    delivery_id,
    delivery_time_minutes
CASE
    WHEN delivery_time_minutes <= 30 THEN 'Fast'
    WHEN delivery_time_minutes <= 60 THEN 'On Time'
    ELSE 'Late'
END AS performance
FROM deliveries;
```

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delivery_id	delivery_time_minutes	Performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late.

The tickets table

Question 6

```
SELECT
    issue_type,
    Priority
CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'
    END AS priority_label
FROM tickets;
```

issue_type	Priority	Priority_label
Login issue	1	Low
Server down	3	High
Slow System	2	Medium
Email error	2	Medium
Password reset	1	Low

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The attendance table Question7

```
SELECT
    student_id,
    (day_present * 100.0 / total_days) AS attendance_percentage,
CASE
    WHEN (day_present * 100.00 / total_days) >= 90
    THEN 'Excellent'
    WHEN (days_present * 100.00 / total_days) >= 75
    THEN 'Good'
    ELSE 'Needs Improvement'
END AS attendance_status

FROM attendance;
```

Student_id	attendance_percentage	attendance_status
1	90.00	Excellent
2	60.00	Needs improvement
3	96.00	Excellent
4	50.00	Needs Improvement
5	100.00	Excellent

The products_inventory table Question8

```
SELECT
    product_id,
    stock_qty
CASE  WHEN stock_qty = 0 THEN 'out of stock'
      WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low stock'
      WHEN stock_qty > 5 THEN 'in stock'
END AS stock_status
FROM products_inventory.
```

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Product_id	Stock_qty	Stock status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock.

The classes table

Question 9

```
SELECT
    subject,
    enrolled_students
CASE
    WHEN enrolled_students >= 25 THEN 'Large'
    WHEN enrolled_students >= 10 THEN 'Medium'
    ELSE 'Small'
END AS class_size_category
FROM classes;
```

Expected output

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium.

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The payments table Question 10

```
SELECT
    payment_id,
    payment_method,
    amount,
    CASE
        WHEN payment_method = 'cash' AND amount >= 200
        THEN 'Eligible for discount'
        ELSE 'Not Eligible'
    END AS discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50	Not Eligible
2	Cash	200	Eligible for discount
3	Card	150	Not Eligible
4	PayPal	75	Not Eligible
5	Cash	300	Eligible for discount

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